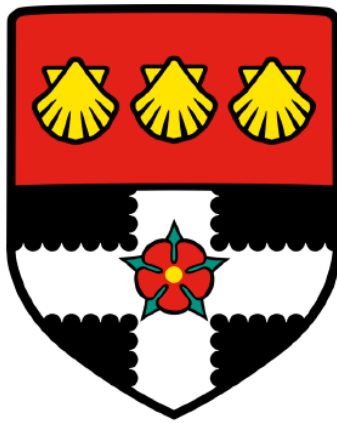


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DEPARTMENT OF METEOROLOGY



Improving Surface Gravity Wave Forecasts Using Data Assimilation

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*A thesis submitted in partial fulfillment of the requirements for the degree of
Master of Science in Atmosphere, Ocean and Climate*

August 2024

Declaration of Authorship

I, Owen Aljabar, declare that this thesis titled, “Improving Surface Gravity Wave Forecasts Using Data Assimilation” and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a research degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

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Abstract

Faculty Name

Department of Meteorology

Master of Science

Improving Surface Gravity Wave Forecasts Using Data Assimilation

by Owen Aljabar

The Thesis Abstract is written here (and usually kept to just this page). The page is kept centered vertically so can expand into the blank space above the title too...

Acknowledgements

The acknowledgments and the people to thank go here, don't forget to include your project advisor. . .

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List of Abbreviations

DA	Data Assimilation
WW3	WAVEWATCH III
LETKF	Local Ensemble Transform Kalman Filter

Chapter 1

Introduction

1.1 The need for Reliable Wave Forecasting

Offshore operations form an important of many industries such as energy and research, and contribute to the economies of coastal countries round the world. Ocean waves pose a significant risk to offshore operations as they can lead to the loss of life and damage to equipment. This means that accurate wave predictions are a critical safety issue for offshore operations, the the operations can only be carried out safely when the waves are sufficiently calm. To provide these predictions the MetOcean Forecasting division of Fugro produces wave forecasts for a client undertaking offshore operations in the Gulf of Mexico. It is important to understand the risk posed by waves in this region since increased cyclone intensity is leading to increased extreme wave heights in the Gulf of Mexico (**Appendini2014**). To avoid unnecessary downtime, they produce ensemble forecasts and apply corrections using a proprietary method based on the alpha factor method described in **Wu2021**.

This proprietary method is referred to by Fugro as a site-specific alpha factor and works by...

As well as these forecasts, Fugro has a buoy based near to the client which produces hourly observations of the sea state. Data assimilation refers to a range of techniques used to integrate observations into a forecast, with the aim of making the forecast more accurate. In this project, we explore the potential of DA to reduce uncertainty and error in these wave forecasts.

1.2 Previous work on DA for Wave Forecasts

Houghton et al., **2023** assimilated wave height observations from satellite altimeters and a global network of drifting Sofar spotter buoys. They compared two different DA methods, a deterministic optimal interpolation (OI) method (cite) and an ensemble-based local ensemble transform Kalman filter (LETKF) (cite). They found that both methods reduced forecast error out to 2.5 days leadtime, and the LETKF gave more physically realistic output states. In this project we will assimilate observations of both wave height and period in the hope that correlations between these variables might give better predictions of both. We will use the ensemble transform Kalman filter (ETKF).

Houghton et al., **2022** used OI to compare the effects of assimilating wave height and spectra. They found that both variables give very similar improvements in predictions of wave height, while spectral assimilation gives better results for other quantities such as period and direction. They also found that the spectral DA did particularly well

FIGURE 1.1: RMSE from spectral DA and wave height (Hs) DA, normalised by the RMSE without DA. Results shown for significant wave height Hs, peak period T_p , mean period T_m , directional spread σ_θ , peak direction D_p and mean direction D_m Figure 3 from Houghton et al. (2022)

in swell-dominated fields when compared to assimilating wave height. A plot showing the RMSE from both approaches is shown in figure 1.1

Caires, Kim, and Groeneweg, 2018 investigated the effect on wave forecasts of using an ensemble Kalman filter (EnKF) method to adjust the wind fields used as forcing in a wave model. They found...

All the studies mentioned forecasted over a wide area (check). A problem with this approach is the inherent difficulty present in observing the ocean which means that observations are sparse, generally coming from drifting buoys which give irregular observations (reword) and satellite altimetry which can be unreliable (check). This is problematic for DA, which relies on regular and frequent observations to work well.

In this project we will be focusing mainly on a particular point of interest and analysing the effectiveness of DA to reduce forecast error at this point. The observations used in this project are also point-based, coming from a fixed buoy near the point of interest (how far?) which measures the sea state hourly. The point, which will be referred to as the buoy location, is of particular interest to a client of Fugro and is located near the Southern coast of the Gulf of Mexico. Its exact location cannot be disclosed due to confidentiality requirements. We hope that the regularity of observations and proximity of the buoy to the point of interest will lead to improved DA performance compared to area-based forecasts.

Another major difference between this study and others that should be noted is the difference in scale: all studies mentioned used many more observations than are used here, and

1.3 Aims of the Project

The main research question in this project is “**What kind of reduction in forecast error can be achieved by assimilating Hs, Tm01 and Tm02 using the ensemble transform Kalman filter method?**”

To this end, we use the ensemble transform Kalman filter (ETKF) to assimilate observations of the three variables and examine the errors resulting from this as compared to the errors before any DA is applied. Measures of error that are examined are:

- The root mean square error (RMSE) of the ensemble mean. We expect the DA to move the ensemble mean closer to the true state so RMSE will be used to quantify and test this. The dependence of RMSE on forecast leadtime and location within the domain will be examined.
- The correlation between ensemble mean and the true state. DA should make the ensemble mean more closely correlated with the true state, so we will test this hypothesis. Correlations in space and in time will both be presented.
- Errors in maximum Hs. Extremes of Hs are of particular interest in these forecasts as these are the events which cause danger and require the pausing of

offshore operations. We will examine the effectiveness of DA to reduce error in the timings and magnitudes of wave height maxima.

Since accuracy in forecasting Hs is safety critical for offshore operations, we consider it to be the most important variable of the three and so a separate DA run is performed where only Hs is assimilated. Results from this run are compared to results from assimilating the three variables Hs, Tm01 and Tm02 together.

Chapter 2

Theory

This chapter outlines the theoretical aspects of this project, covering both the linear wave theory used in the wave model and the theory of the data assimilation methods used. The aim of the chapter is to describe all the theory required to understand the methods used in this project.

2.1 Ocean Waves

Ocean waves are a distinctive visual feature of the ocean surface, and very important to offshore operations such as wind farms. This section will describe some of the physical processes influencing ocean waves, as well as the theory of linear waves and wave energy spectra as modelled by the WAVEWATCH III (WW3) model used in this project.

2.1.1 Physical Processes

Ocean waves are an instance of surface gravity waves, meaning they propagate on the interface between two fluids of different densities. Their restoring force is gravity. The focus of this project is on waves with periods from 10s - 0.25s, which are generated by wind, although ocean waves can exist on a wide range of scales and be influenced by other factors such as surface tension and local air pressure variations (**Holthuijsen2007**, pp. 3-7).

There are multiple hypothesised physical mechanisms for the generation of waves by wind (e.g. ...). The approach from Miles is popular, and described in detail in **Janssen2004**, chapter 3. Qualitatively speaking, Miles hypothesises that waves are generated by instability induced by the velocity shear between the air and ocean. Wind-generated waves will propagate in the same direction as the wind that generated them and will be of larger amplitude when the wind is stronger. (check)

The wind almost always acts as a source of wave energy, so there has to be a process present to dissipate waves and act as an energy sink. These processes are poorly understood but **Janssen2004**, section 4.9 describes multiple physical mechanisms which may contribute to wave dissipation. One important process affecting dissipation is wave breaking, which occurs in shallow waters near shorelines. Water's molecular viscosity can lead to the damping of shorter waves and larger waves can be damped by the surface eddies induced by wave breaking.

Ocean waves in deep water are dispersive, and waves with longer wavelengths travel faster than waves with shorter wavelengths (equations 2.3-2.5). This leads to a phenomenon known as swell, which is the name given to long waves that have travelled

FIGURE 2.1: Schematic of the generation of wind and swell waves.

far from their source. Due to dispersion these waves have no short-wavelength component, so swell waves have longer wavelengths and periods than waves near the site of generation (known as wind sea). The other feature of swell waves is their regular appearance. While a wind sea has wave components in many different directions, by the time the waves have travelled away from the source these different components will spread out so that they do not overlap, and the resulting swell will be almost unidirectional. See figure 2.1 for a schematic.

2.1.2 Linear Wave Theory

This section will outline the linear wave theory relevant to this project, as derived in Vallis2017, page 253. Linear wave theory is an approximation, and other higher-order theories of surface gravity waves exist, such as Stokes waves (cite) and cnoidal waves (cite). These higher-order theories will not be considered here since the WAVE-WATCH III model used in this project models the ocean surface as a linear superposition of simple linear waves.

Parameters describing linear waves include the wavenumber k , direction θ and frequency ω . The first two describe the wave vector $\mathbf{k}$, which has magnitude k and direction θ i.e.

$$\mathbf{k} = k \cos \theta \hat{\mathbf{i}} + k \sin \theta \hat{\mathbf{j}} \quad (2.1)$$

where $\hat{\mathbf{i}}, \hat{\mathbf{j}}$ are unit vectors pointing East and North respectively. This means that θ is the angle between $\mathbf{k}$ and $\hat{\mathbf{i}}$, measured anticlockwise from $\hat{\mathbf{i}}$. ω here denotes the absolute frequency measured in a fixed frame of reference. If the medium is subject to currents, this will differ from the relative frequency σ measured in a frame of reference moving with the mean current, due to the Doppler effect. This difference is quantified by the equation

$$\omega = \sigma + \mathbf{k} \cdot \mathbf{U} \quad (2.2)$$

and σ and k are related by the dispersion relation

$$\sigma^2 = gk \tanh kD \quad (2.3)$$

The dispersion relation has two limiting forms:

- Deep water waves, where $kD \gg 1$, so $\tanh kD \approx 1$ and $\sigma^2 \approx gk$.
- Shallow water waves, where $kD \ll 1$, so $\tanh kD \approx kD$ and $\sigma^2 \approx gk^2D$

The phase speed is then given by

$$c = \frac{\sigma}{k} \approx \begin{cases} \sqrt{\frac{g}{k}} & \text{in deep water} \\ \sqrt{gD} & \text{in shallow water} \end{cases} \quad (2.4)$$

and the group speed

Explain the physical relevance of the group speed vs the phase speed

may be expressed as

$$c_g = \frac{\partial \sigma}{\partial k} = \frac{1}{2}c \left(1 + \frac{2kD}{\sinh 2kD} \right) \approx \begin{cases} \frac{1}{2}c & \text{in deep water} \\ c & \text{in shallow water} \end{cases} \quad (2.5)$$

The phase speed is the speed at which the wave peaks and troughs move, whereas the group speed is the speed at which wave groups move. The group speed is then the speed at which energy is transported by the wave. The wave travels in the direction of the wave vector $\mathbf{k}$ so the group and phase velocities are given by

$$\begin{aligned} \mathbf{c} &= c \cos \theta \hat{\mathbf{i}} + c \sin \theta \hat{\mathbf{j}} \\ \mathbf{c}_g &= c_g \cos \theta \hat{\mathbf{i}} + c_g \sin \theta \hat{\mathbf{j}} \end{aligned}$$

We can now use scale analysis to see that ω and σ generally have similar values. The ratio between $\mathbf{k} \cdot \mathbf{U}$ and σ in equation 2.2 is

$$\frac{\mathbf{k} \cdot \mathbf{U}}{\sigma} \sim \frac{kU}{\sigma} = \frac{U}{c}$$

Where $U = |\mathbf{U}|$ is the mean current speed. Since $c \gg U$ in general, this ratio is small and so $\omega \approx \sigma$.

2.1.3 Wave Spectra

Note that equations 2.2 - 2.5 are valid for linear waves of a single frequency and direction. Real ocean waves are more complicated than this, containing multiple frequencies propagating in multiple directions. The WAVEWATCH III model used in this project models ocean waves as a superposition of these simple linear waves.

Surface waves with multiple frequencies and directions are represented using a wave spectrum $F = F(\sigma, \theta)$, which is also a function of position and time. F represents the sea surface height variance associated with a particular frequency σ and direction θ , so the total height variance is

$$\overline{\eta^2} = \iint F d\sigma d\theta \quad (2.6)$$

This is related to the wave energy via $E = \rho g \overline{\eta^2}$. A rigorous definition of F is given in **Holthuijsen2007**, section 3.5.6. One can also define the unidirectional spectrum $S(\sigma) = \int F(\sigma, \theta) d\theta$ which quantifies the surface height variance as a function of frequency. This can be integrated to calculate the energy contained in a specific frequency band $\Delta\sigma$:

$$\Delta E = \int_{\Delta\sigma} S(\sigma) d\sigma \quad (2.7)$$

FIGURE 2.2: An example unidirectional spectrum from a point in the gulf of Mexico. Calculated from WW3 spectral forecast data.

S is a helpful diagnostic from which information about the appearance of the waves can be inferred. A widely spread spectrum implies irregular waves with many different frequency components whereas a tight spectrum implies regular waves with one dominant frequency. Low frequency waves correspond to swell, while high frequency waves correspond to a wind sea. An example of a unidirectional spectrum is plotted in figure 2.2.

Here, we also define the wave spectral moments which are derived from the unidirectional spectrum:

$$m_k = \int S(\sigma) \sigma^k d\sigma, m = \dots, -1, 0, 1, \dots \quad (2.8)$$

Note that $m_0 = \overline{\eta^2}$ is the overall surface height variance. These moments can be used to derive other important wave variables from the spectrum, such as significant wave height and wave period. In this project we focus on the significant wave height H_s and two measures of frequency Tm01, Tm02. These variables are defined below in equation 2.11 (expand?).

$$H_s = 4\sqrt{m_0} \quad (2.9)$$

$$Tm01 = \frac{m_0}{m_1} \quad (2.10)$$

$$Tm02 = \sqrt{\frac{m_0}{m_2}} \quad (2.11)$$

Finally we define the wave action

$$N(\sigma, \theta) = \frac{F(\sigma, \theta)}{\sigma}$$

N is conserved in the absence of sources and sinks (**Bretherton1968**), so it very useful for wave modelling. WW3 works by evolving the wave action according to the action of sources and sinks (see section 3.1).

As well as the model, we also have some direct observations of the waves which we would like to make use of to improve our predictions. We use techniques of data assimilation to achieve this.

Add general explanation what DA is doing. I.e. finding estimates that lie closer to truth using the information in the observations.

2.1.4 Vector Representations and Cost Function

2.1.5 Vector Representations and Cost Function

We take a vector $\mathbf{x} \in \mathbb{R}^n$ to represent the state of the system, assuming it has a normal background distribution $\mathbf{x} \sim N(\mathbf{x}_b, \mathbf{B})$. We take a vector $\mathbf{y} \in \mathbb{R}^p$ to represent a set of

observations, and assume $\mathbf{y}$ is related to $\mathbf{x}$ via an observation operator $\mathcal{H} : \mathbb{R}^n \rightarrow \mathbb{R}^p$:

It might be better to introduce $\mathbf{x}_{true}$ here to make a distinction between a specific value of the state and reference to the state in general.

$$\mathbf{y} = \mathcal{H}(\mathbf{x}) + \varepsilon$$

where the error term ε is assumed to be normally distributed with mean zero and covariance $\mathbf{R}$: $\varepsilon \sim N(0, \mathbf{R})$. We call the vector spaces containing $\mathbf{x}$ and $\mathbf{y}$ the state space and observation space respectively. In operational contexts, the dimensions of these spaces are very large: typically $n \sim 10^8$ and $p \sim 10^6$. This means that considerable computational resources are required in most cases and computational efficiency is a key consideration in any operational DA implementation.

In our case $\mathcal{H}$ is linear i.e. there is a matrix $\mathbf{H} \in \mathbb{R}^{n \times p}$ such that $\mathcal{H}(\mathbf{x}) = \mathbf{H}\mathbf{x}$, and so $\mathbb{P}(\mathbf{y}|\mathbf{x}) = N(\mathbf{H}\mathbf{x}, \mathbf{R})$. We can then express the distribution of $\mathbf{x}$ given some set of observations $\mathbf{y}$ using Bayes' theorem:

$$\mathbb{P}(\mathbf{x}|\mathbf{y}) = \mathbb{P}(\mathbf{y}|\mathbf{x}) \frac{\mathbb{P}(\mathbf{x})}{\mathbb{P}(\mathbf{y})} \quad (2.12)$$

This allows $\mathbb{P}(\mathbf{x}|\mathbf{y})$ to be determined since $\mathbb{P}(\mathbf{y}|\mathbf{x})$ and $\mathbb{P}(\mathbf{x})$ are known, and $\mathbb{P}(\mathbf{y})$ is independent of $\mathbf{x}$ so it acts as a normalisation constant. Since all the distributions involved are normal, $\mathbb{P}(\mathbf{x}|\mathbf{y})$ will be normally distributed also:

You need define $\sim$ here.

$$\mathbb{P}(\mathbf{x}|\mathbf{y}) \sim e^{-\frac{1}{2}(\|\mathbf{x} - \mathbf{x}_b\|_{\mathbf{B}^{-1}}^2 + \|\mathbf{y} - \mathbf{H}\mathbf{x}\|_{\mathbf{R}^{-1}}^2)} \quad (2.13)$$

adopting the notation $\|\mathbf{a}\|_{\mathbf{M}}^2 = \mathbf{a}^\top \mathbf{M} \mathbf{a}$. Finding the most likely value of $\mathbf{x}$ given this distribution is equivalent to minimising a cost function

$$J(\mathbf{x}) = \frac{1}{2} \|\mathbf{x} - \mathbf{x}_b\|_{\mathbf{B}^{-1}}^2 + \frac{1}{2} \|\mathbf{y} - \mathbf{H}\mathbf{x}\|_{\mathbf{R}^{-1}}^2 \quad (2.14)$$

The first term of 2.14 represents the deviation of $\mathbf{x}$ from the background state and the second term represents the deviation from the observations, with the overall cost function being a sum of these two weighted by the background and observation error matrices. This means that when the background error is large compared to the observational error, the observations will be given more weight and vice versa.

2.1.6 Kalman Filters

The state that minimises this cost function is called the analysis $\mathbf{x}_a$ and may be expressed as

$$\mathbf{x}_a = \mathbf{x}_b + \mathbf{K} \mathbf{d}_b^o \quad (2.15)$$

where $\mathbf{d}_b^o = \mathbf{y} - \mathbf{H}\mathbf{x}$ is the innovation and $\mathbf{K} = \mathbf{B}\mathbf{H}^\top (\mathbf{H}\mathbf{B}\mathbf{H}^\top + \mathbf{R})^{-1}$ is the Kalman gain (derived in **Asch2016**, pp. 92-96). Note that the increment $\mathbf{d}_b^a = \mathbf{x}_a - \mathbf{x}_b$ is given by $\mathbf{B} \left(\mathbf{H}^\top (\mathbf{H}\mathbf{B}\mathbf{H}^\top + \mathbf{R})^{-1} \mathbf{d}_b^o \right)$, so it always lies in the column space of $\mathbf{B}$. This means that $\mathbf{B}$ is crucial to determining how information from the observations is spread around the domain by the DA system. In particular, if $\mathbf{B}$ is rank deficient, then all possible increments lie within a strict subspace of the state space. This reduces the degrees of freedom in the corrections the DA system can make, which limits the performance of the system.

DA methods which use equation 2.15 are called Kalman filters, in which context it is called the analysis step. Kalman filters generally alternate this analysis step with a forecast step in which a forecast model is used to evolve the state forward to the next observation time. In this project we do not have access to the forecast model, so we will only make use of the analysis step to update the state.

The specific method we will use is called the ensemble transform Kalman filter (ETKF) in which the analysis is performed in ensemble subspace rather than in the state space. Algebraically this means expressing the increment as a linear combination of ensemble perturbations

$$\mathbf{X}_f = \frac{1}{\sqrt{N-1}} \left(\mathbf{x}_1^f - \bar{\mathbf{x}}^f, \mathbf{x}_2^f - \bar{\mathbf{x}}^f, \dots, \mathbf{x}_N^f - \bar{\mathbf{x}}^f \right) \quad (2.16)$$

$$\mathbf{x}^a = \bar{\mathbf{x}}^f + \mathbf{X}_f \mathbf{w}^a \quad (2.17)$$

Where $\mathbf{x}_i^f$ is the state of ensemble member i prior to assimilation and the ensemble mean $\bar{\mathbf{x}}^f$ is used as the background state. From **Asch2016** pp.163-164, the optimal coefficient vector $\mathbf{w}^a$ is given by

$$\mathbf{w}^a = \left(\mathbf{I}_N + \mathbf{X}_f^\top \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H} \mathbf{X}_f \right)^{-1} \mathbf{X}_f^\top \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{d}_b^o \quad (2.18)$$

This method is algebraically equivalent to working in state space with the background error covariance matrix given by the forecast error covariance matrix

$$\mathbf{P}_f = \mathbf{X}_f \mathbf{X}_f^\top \quad (2.19)$$

This matrix is rank deficient since its column space has dimension equal to the ensemble size, but working in ensemble space has considerable advantages over working in state space in terms of computational cost (**Hunt2007**). The full analysis ensemble can then be generated from the analysis state derived in equation 2.18 via

$$\mathbf{x}_i^a = \mathbf{x}^a + \dots \quad (2.20)$$

And the analysis error covariance $\mathbf{P}_a$ can be defined similarly to $\mathbf{P}_f$ but from the analysis ensemble:

$$\mathbf{X}_a = \frac{1}{\sqrt{N-1}} (\mathbf{x}_1^a - \mathbf{x}^a, \mathbf{x}_2^a - \mathbf{x}^a, \dots, \mathbf{x}_N^a - \mathbf{x}^a) \quad (2.21)$$

$$\mathbf{P}_a = \mathbf{X}_a \mathbf{X}_a^\top \quad (2.22)$$

Another advantage of the ETKF (compared to the stochastic ensemble Kalman filter) is that $\mathbf{P}_a$ matches exactly the form given by Kalman theory (cite?)

2.1.7 Practical Aspects

When calculating $\mathbf{P}^f$ from the ensemble, spurious covariances may arise at long distances, especially when the ensemble size is small. This is problematic as it can lead to large increments far from observation locations, which is physically unrealistic. One way to fix this is called localisation, and involves multiplying each component of the background error matrix $\mathbf{P}_{ij}^f$ by a factor which depends on the distance between gridpoints i and j . Representing this distance with $d(i, j)$, we have the following equation:

$$\mathbf{P}_{ij}^{f'} = \mathbf{P}_{ij}^f f(d(i, j)) \quad (2.23)$$

The function f should be close to 1 when $d(i, j)$ is small and close to 0 when $d(i, j)$ is large, to suppress covariances at long distances while allowing them to exist at short distances (FIND SOURCE)

Another problem that may be encountered is... (filter divergence - tailor to our case)

PROOF OF EQUIVALENCE WITH WEAK 4DENVAR

EnKF description - may not be needed

In most contexts, observations are not all simultaneous but instead come from a sequence of times $t = 0, 1, \dots$. One approach to assimilating non-simultaneous observations is to use a filter method such as the Ensemble Kalman Filter (Anderson, 2001; Evensen, 1994; Houtekamer and Mitchell, 1998). A filter method consists of successive analysis steps, in which the observations from time t are assimilated according to equation 2.15, alternating with update steps, in which the forecast model is used to evolve the state forward to time $t + 1$.

Where N is the ensemble size, $\mathbf{x}_t^{(i)}$ is the state of ensemble member i before the analysis step at time t and $\bar{\mathbf{x}}_t = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_t^{(i)}$ i.e. the ensemble mean.

In general, the observations from each time t may be available at different locations, or observations of different variables may be available, so the observation operator and observation error covariance matrix are functions of time. Denoting these at time t by $\mathbf{H}_t, \mathbf{R}_t$, the Kalman gain matrix takes the form

$$\mathbf{K}_t = \mathbf{P}_t^f \mathbf{H}_t^\top \left(\mathbf{H}_t \mathbf{P}_t^f \mathbf{H}_t^\top + \mathbf{R}_t \right)^{-1}$$

The EnKF method is prone to a problem called filter divergence - sampling error in the calculation of $\mathbf{P}_t^f$ means the magnitude of the forecast error covariance tends to erroneously shrink over time (Jazwinski, 1970). This means that after some time, the ensemble distribution becomes very tight and further observations do not affect

the state of the ensemble, resulting in forecast analysis states which diverge from the true state. (schematic) To counteract this, we will employ the method of inflation described in Anderson and Anderson, 1999, where the forecast error covariance matrix is multiplied by a factor α before the analysis step is performed.

2.1.8 (Description of 4DEnVar - may not be needed)

In our case, we have observations from multiple times $t = 0, 1, \dots, T$ spread out over some time window, and we expect the state $\mathbf{x}$ to evolve over this time window. This evolution may be predicted using the model as follows: let $\mathbf{x}_0$ represent the state of the system at $t = 0$, then we define the model operator $\mathcal{M}_t$ for each t such that $\mathbf{x}_{t+1} = \mathcal{M}_t(\mathbf{x}_t)$ as predicted by the model. The cost function then becomes

$$J(\mathbf{x}_0) = \frac{1}{2} \|\mathbf{x}_0 - \mathbf{x}_b\|_{\mathbf{B}_0^{-1}}^2 + \frac{1}{2} \sum_{t=0}^T \|\mathbf{y}_t - \mathbf{H}_t \mathbf{x}_t\|_{\mathbf{R}_t^{-1}}^2 \quad (2.24)$$

$$= \frac{1}{2} \|\mathbf{x}_0 - \mathbf{x}_b\|_{\mathbf{B}_0^{-1}}^2 + \frac{1}{2} \sum_{t=0}^T \|\mathbf{y}_t - \hat{\mathbf{h}}_t \mathbf{x}_0\|_{\mathbf{R}_t^{-1}}^2 \quad (2.25)$$

where the observations $\mathbf{y}_t$ and matrices $\mathbf{H}_t, \mathbf{R}_t$ are valid at time t , and

$\hat{\mathbf{h}}_t$ is an operator and should therefore be capitalized

$\hat{\mathbf{h}}_t = \mathbf{H}_t \mathcal{M}_{0 \rightarrow t}$ using the abbreviation $\mathcal{M}_{0 \rightarrow t} = \mathcal{M}_{t-1} \mathcal{M}_{t-2} \dots \mathcal{M}_0$.

We will use the 4DEnVar method, where a control variable transform is used to write $\mathbf{x}_0$ as a linear combination of the ensemble perturbations:

$$\begin{aligned} \bar{\mathbf{x}}_b &= \frac{1}{N} \sum_{n=1}^N \mathbf{x}_b^{(n)} \\ \mathbf{X}'_b &= \frac{1}{\sqrt{N-1}} \left(\mathbf{x}_b^{(1)} - \bar{\mathbf{x}}_b, \mathbf{x}_b^{(2)} - \bar{\mathbf{x}}_b, \dots, \mathbf{x}_b^{(N)} - \bar{\mathbf{x}}_b \right) \\ \mathbf{x}_0 &= \bar{\mathbf{x}}_b + \mathbf{X}'_b \chi \end{aligned}$$

Here $\mathbf{x}_b^{(n)}$ represents the background state from ensemble member n , and N is the ensemble size. The cost function can then be rewritten in terms of χ and simplified by linearising $\hat{\mathbf{h}}_t$ around $\bar{\mathbf{x}}_b$:

$$J(\chi) = \frac{1}{2} \|\chi\|^2 + \frac{1}{2} \sum_{t=0}^T \left\| \mathbf{y}_t - \hat{\mathbf{h}}_t \bar{\mathbf{x}}_b - \hat{\mathbf{H}}_t \mathbf{X}'_b \chi \right\|_{\mathbf{R}_t^{-1}}^2 \quad (2.26)$$

where $\hat{\mathbf{H}}_t = \mathbf{H}_t \mathbf{M}_{0 \rightarrow t}$ is the linearisation of $\hat{\mathbf{h}}_t$. This cost function is simplified by taking a background error covariance matrix of the form $\mathbf{B} = \mathbf{X}'_b \mathbf{X}'_b{}^T$, which reduces to the identity under the control variable transform. This cost function has gradient

$$\nabla J(\chi) = \chi + \sum_{t=0}^T \left(\hat{\mathbf{H}}_t \mathbf{X}'_b \right)^\top \mathbf{R}_t^{-1} \left(\hat{\mathbf{h}}_t \mathbf{X}'_b \chi + \hat{\mathbf{H}}_t \bar{\mathbf{x}}_b - \mathbf{y}_t \right)$$

A major advantage of this formulation is that we can make the approximation

$$\mathbf{M}_{0 \rightarrow t} \left(\mathbf{x}_b^{(n)} - \bar{\mathbf{x}}_b \right) \approx \mathcal{M}_{0 \rightarrow t} \left(\mathbf{x}_b^{(n)} \right) - \overline{\mathcal{M}_{0 \rightarrow t} \left(\mathbf{x}_b \right)}$$

Since $\mathbf{H}_t \mathbf{M}_{0 \rightarrow t} \left(\mathbf{x}_b^{(n)} - \bar{\mathbf{x}}_b \right)$ is the n^{th} column of $\hat{\mathbf{H}}_t \mathbf{X}'_b$, this approximation allows us to compute the cost function and its gradient without a tangent linear or adjoint model, which is a computationally expensive aspect of most variational data assimilation schemes (Bannister, 2017).

Chapter 3

Experimental Setup

This chapter is intended to describe the details of the experimental setup used in this project. It contains a description of the wave model used to generate the forecasts, as well as an overview of the background state, the observations used and the errors present in these observations. Finally a description is given of the twin experiment method used in this project, as well as the data assimilation setup.

3.1 Model Description and Domain

Waves are modelled by Fugro using WW3 (**WW3DG2019**), which uses as its base spectrum a wavenumber-direction spectrum $F(k, \theta)$, rather than the frequency-direction spectrum described in section 2.1.3. The spectrum is discretised using 29 sampled wavenumbers k and 24 angles θ . The wavenumber-direction spectrum is related to the frequency-direction spectrum by the Jacobian transform

$$F(\sigma, \theta) = \frac{\partial k}{\partial \sigma} F(k, \theta) = \frac{1}{c_g} F(k, \theta)$$

Wave propagation is then described by the equation

$$\frac{D}{Dt} N(k, \theta) = \frac{S}{\sigma} \quad (3.1)$$

Where $N(k, \theta) = F(k, \theta)/\sigma$ as before and S describes the net effect of sources and sinks for the spectrum $F(k, \theta)$.

In deep water, S consists of three main parts: an atmospheric interaction term S_{in} , a nonlinear wave-wave interaction term S_{nl} and a dissipation term S_{ds} .

S_{in} is dominated by an exponential wind-wave growth term, so S_{in} is equal to some coefficient multiplied by $\sigma N(k, \theta)$. This leads to a feedback which is usually positive but can be negative in the case of swell, and represents the main source of wave energy.

S_{ds} is generally dominated by wave breaking, and it represents a sink of wave energy. Similar to S_{in} , it is equal to a negative coefficient multiplied by $\sigma N(k, \theta)$. The balance between S_{in} and S_{ds} determines the overall growth or decay of wave energy.

S_{nl} represents interactions between waves due to nonlinear processes, which results in energy transfer between waves of different frequencies. (maybe expand - **Janssen2004**)



FIGURE 3.1: Model domains shown on map.

In shallow water, processes such as wave-bottom interactions, wave breaking and reflection off shorelines become important, so additional source terms are included to describe this.

From the forecasted spectra, other variables such as wave height, period and mean direction are derived. The three variables we will be focusing on in this study are significant wave height H_s , and two measures of the dominant wave period T_{m01} and T_{m02} . These are related to the spectrum via the spectral moments (equations 2.8, 2.11).

The reason for assimilating both T_{m01} and T_{m02} is that they are defined from different spectral moments, so assimilating both will provide more information about the spectrum than either one alone, which should lead to better DA results.

As forcing, WW3 uses just the 10 metre wind, and this comes from the GEFS ensemble. The GEFS forecast contains one control run plus 20 perturbed ensemble members. Each GEFS run acts as forcing for one member of the wave ensemble. (Do they?)

The domain in which forecasts are produced lies between $15.0 - 33.0^\circ\text{N}$ and $79.1 - 100^\circ\text{W}$ in the gulf of Mexico. Fugro use a nested domain to produce the forecasts, running WW3 on a 0.5° resolution globally rectangular grid, and two higher resolution nested grids covering the region of interest. There is a 0.1° resolution grid spanning the region described above, and a 0.05° grid spanning from $18 - 22.5^\circ\text{N}$ and $90.5 - 96.5^\circ\text{W}$. These domains are shown on a map in figure 3.1.

Forecasts are produced at 12Z each day, and run up to seven days ahead. We have seven of these daily runs available, with start dates spanning from 06/03/2022 to 12/03/2022. A schematic of this process is shown in figure 3.2. As we do not have access to the model code in this project, we will treat each of these runs as one 'timestep' of the DA system. Further description of how this works can be found in section 3.5.

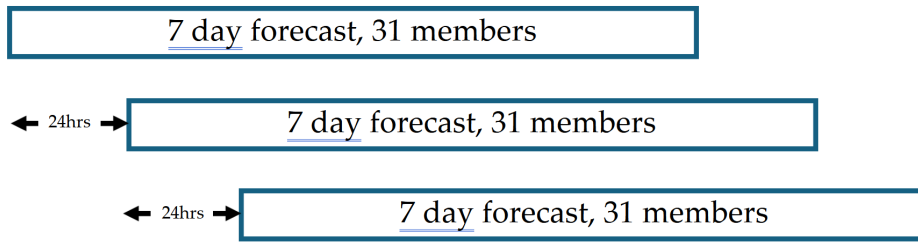


FIGURE 3.2: Schematic of the forecasting setup used by Fugro. A new 7-day 31-member ensemble forecast is produced at 12Z each day.

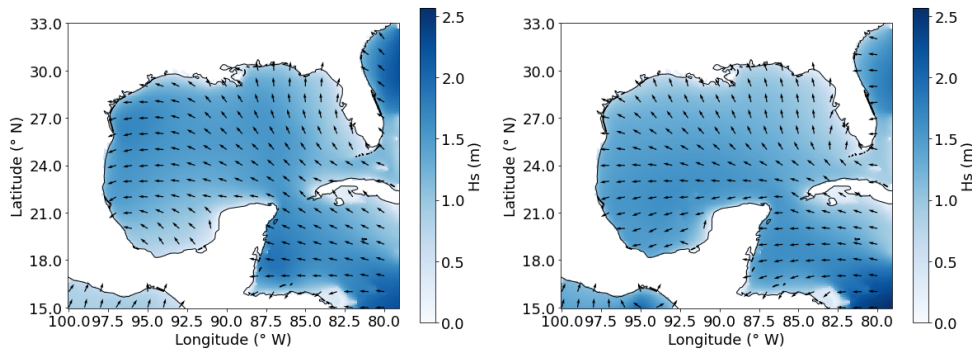


FIGURE 3.3: Left: Hs (colours) and wave direction (arrows) averaged between 12Z 06/03/2022 and 12Z 12/03/2022. Right: The same averaged between 12Z 13/03/2022 and 12Z 19/03/2022

3.2 Background State and Climatology

In this section we present a general overview of the background state and climatology in the domain. This will mostly come from the forecast data which spans a period from 06/03/2022 to 19/03/2022. Since the available observations span a longer period of roughly six months, observations will also be used to identify broader climatological features at the buoy location.

3.2.1 Model Background State

Figure 3.3 shows the significant wave height and direction averaged over the first six days and last six days of the forecast period. Hs is mostly between 1m and 2m in the central gulf, with lower wave heights of < 1m in sheltered coastal areas such as the western side of Florida and the Yucatan peninsula. The central part of the domain has largely east-southeasterly waves over both periods, which diverge after they pass between Yucatan and Cuba. The two periods are separated by the passage of strong cold front (figure 3.4) and there is a slight shift to more easterly waves after the front has passed. This change is likely related to the wind backing associated with the front. The southern part of the GoM has slightly higher waves, around 1 - 1.5m, in the six day period after the passage of the front. (check other climatologies).

There is a large peak in Hs on the 12/03/2022, related to a cold front passing through the region. A map of Hs in the GoM at 18:00 on 06/03/2022 is shown in figure 3.4, along with a surface chart from the same time. The cold front is visible on the chart, and high waves are present in a large region behind the front. This is in line with **Appendini2014**, a study into the wave climatology of the GoM. In that paper, cold

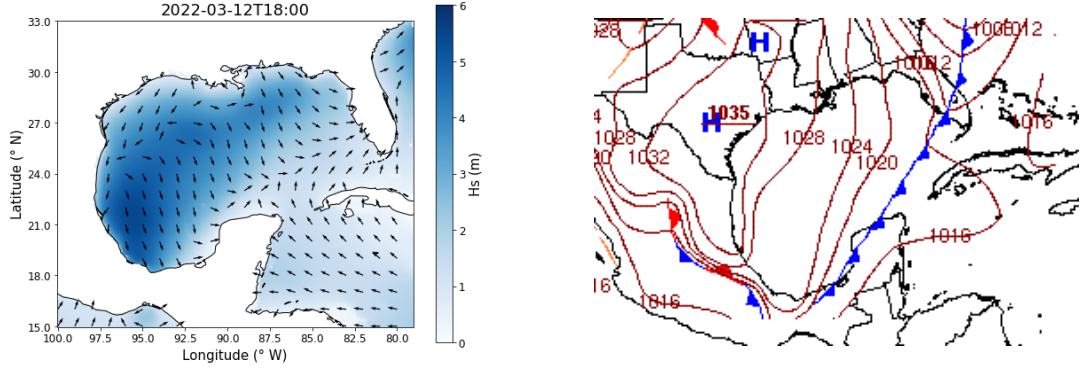


FIGURE 3.4: Left: H_s (colours) and direction (arrows) in the GoM at 18Z on 12/03/2022. Right: Surface analysis chart valid for 18Z on 12/03/2022, taken from NOAA WPC surface analysis archive

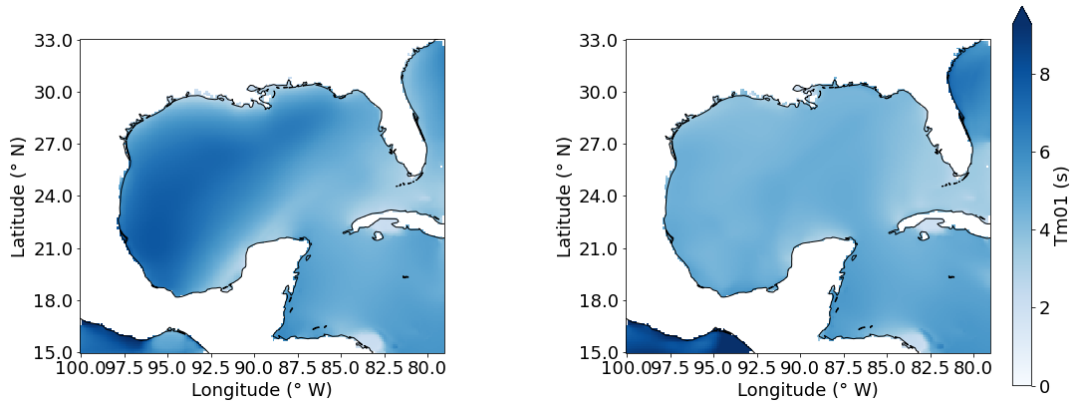


FIGURE 3.5: Left: $Tm01$ at 18Z on 12/03/2022. Right: $Tm01$ averaged over the period from 12Z 10/03/2022 to 12Z 12/03/2022

fronts are identified as a common driver of high waves between October and April, with the most intense peaks occurring between December and March.

Figure 3.5 shows $Tm01$ at 18Z 12/03/2022 and averaged over the preceding 48 hours. The high waves associated with the cold front at 18Z have a period of ~ 8 s, which is higher than the period in the rest of the GoM at that time and also higher than the waves across the entire GoM in the 48 hours prior. This suggests these waves are swell waves.

Figure 3.6 shows a timeseries of H_s ensemble mean and variance at the buoy location from the forecast starting 12Z 10/03/2022. We see the peak on the 12th as seen in figure 3.4. We also see that the variance is a strong function of the mean. The variance in H_s peaks just before the peak in the mean, and drops back to a lower value as the mean reaches its peak. This is notable since the diagonal elements of our forecast error covariance matrix $\mathbf{P}^f$ are simply the ensemble variances at each gridpoint and timestep, so ensemble variance is a good measure of the magnitude of the background error. This means that the background error in H_s is highest just before the peak in H_s .

Figure 3.7 shows the same as 3.6 but for $Tm01$. Again we see a peak in the variance just before the peak in the mean. In this case peak in $Tm01$ is preceded by a dip, and this dip is also associated with a peak in the ensemble variance. This means that

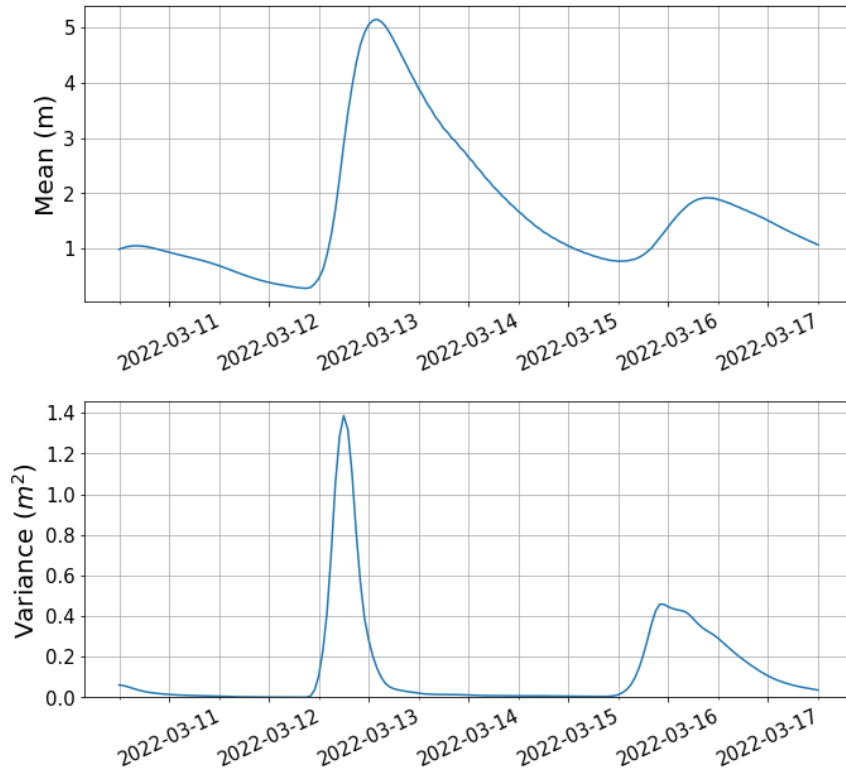


FIGURE 3.6: Timeseries of forecast Hs ensemble mean and variance at the buoy location from the 10/03/2022 model run.

the background error in Tm01 is high for a period of ~ 24 hours, as opposed to ~ 12 hours for Hs.

3.2.2 Climatology at Buoy Location

Figure 3.8 shows a timeseries of Hs and Tm01 as measured by the buoy. We that outside of specific peaks, Hs is generally $< 1\text{m}$. This is consistent with the model background state which showed Hs in the buoy region to be $< 1\text{m}$ for much of the forecast period. The peaks generally consist of a sharp increase in Hs followed by a more gradual decrease down to the baseline level. This is consistent with the slightly higher waves seen in the model background state in the buoy region after the passage of the front. As seen in the background state, the peaks in Hs are associated with peaks in Tm01. This suggests that the high waves seen at the buoy are travelling swell waves and are not generated in the vicinity of the buoy.

Figure 3.9 shows a density plot of observed Hs and wave direction at the buoy location. There is a clear peak in frequency corresponding to waves of height $\sim 0.5\text{m}$ coming from a northeasterly direction. Some higher waves (up to $\sim 5\text{m}$) are present coming from a north-northwesterly direction. This lines up with the direction of waves produced by the cold front on 12/03/2022, suggesting that the waves produced by this kind of event consistently reach the buoy from a north-northwesterly direction. Since the buoy is near the coast in the southern part of the gulf, it is sheltered from the south so there are virtually no waves coming from this direction.

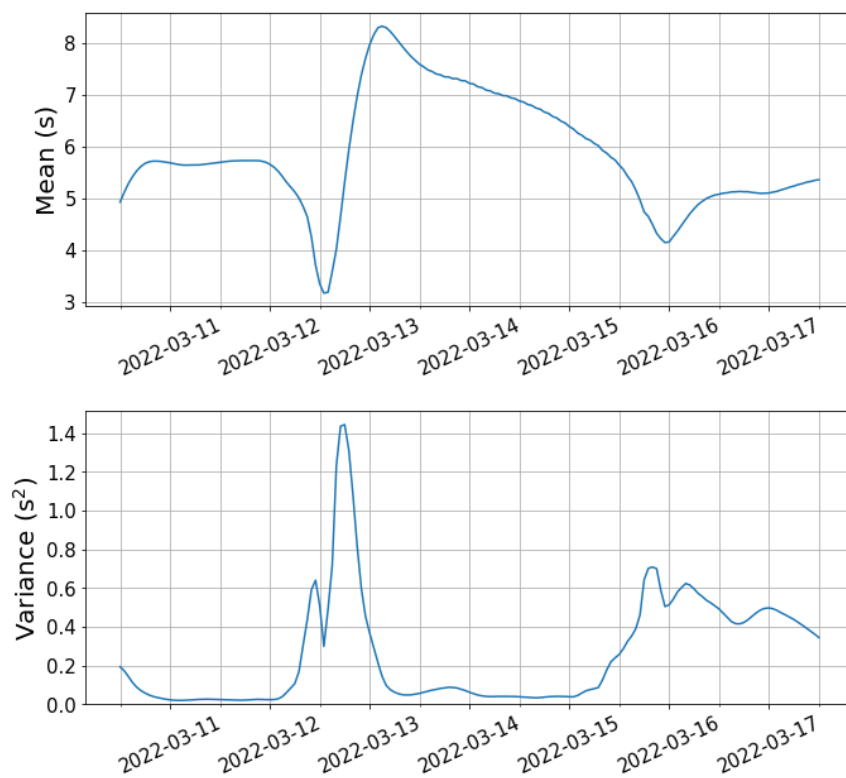


FIGURE 3.7: Timeseries of forecast Tm01 ensemble mean and variance at the buoy location from the 10/03/2022 model run.

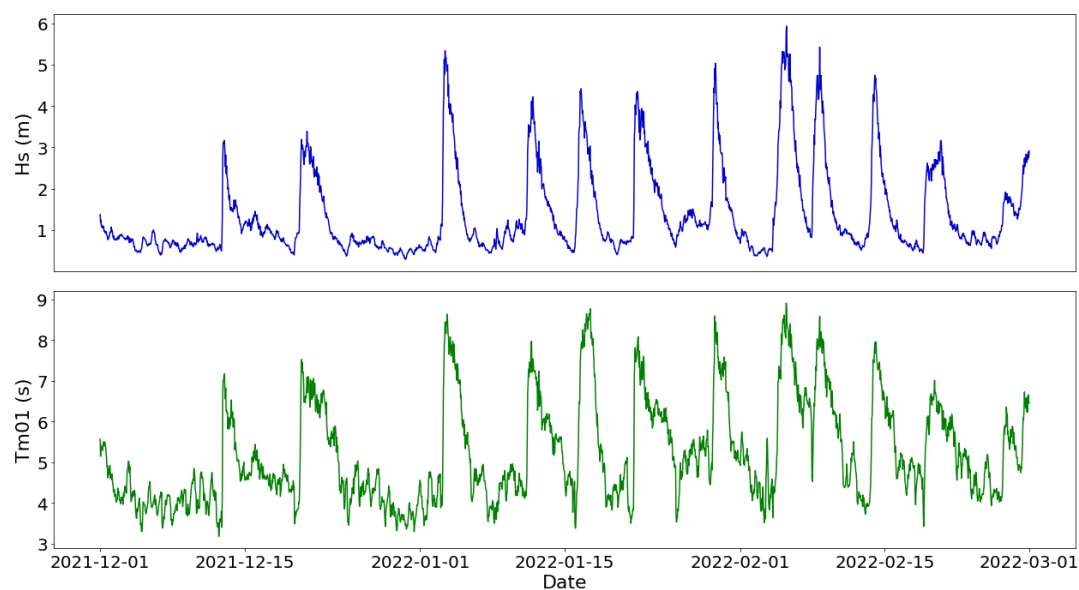


FIGURE 3.8: Timeseries of observed Hs (top) and Tm01 (bottom) over a three-month period from 01/12/2021 to 01/03/2022

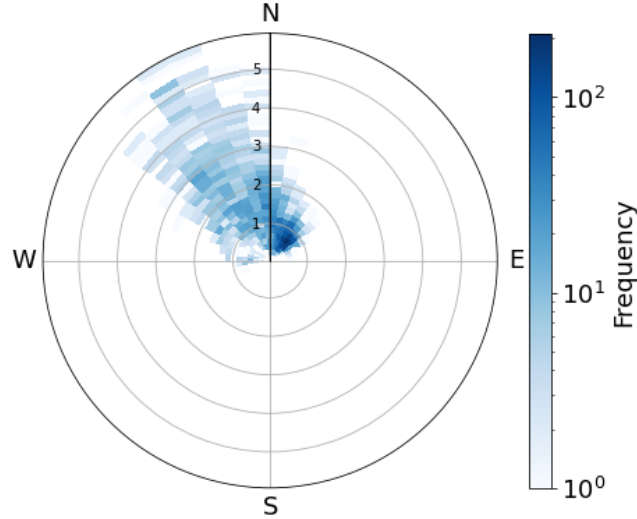


FIGURE 3.9: Density plot of H_s (radial axis) and wave direction at the buoy location.

3.3 ADCP Observations

Our observations come from an acoustic doppler current profiler (ACDP) buoy, which estimates the spectrum by measuring the wave induced velocity and surface height variations. According to **Vallis2017**, the velocity and surface height anomaly induced by a simple wave with wave vector $\mathbf{k} = (k_x, k_y)$, frequency σ and amplitude a is

$$\begin{aligned} u &= a \frac{k_x}{\sigma} g \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \cosh k(z + D) \\ v &= a \frac{k_y}{\sigma} g \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \cosh k(z + D) \\ w &= -ia \frac{k}{\sigma} g \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \sinh k(z + D) \\ \eta &= a \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \end{aligned}$$

where only the real part of each expression should be taken. Note that the horizontal velocity (u, v) is always in the same direction as the wave vector $\mathbf{k}$. From the ADCP measurements, a timeseries of wave induced energy per unit area $E = \int_{-D}^{\eta} \frac{1}{2} \rho_0 \|\mathbf{u}\|^2 dz + \frac{1}{2} \rho_0 \eta^2$ can be calculated. This is equivalent/a good approximation to the wave energy in equation 2.6 (source) so a Fourier transform of E gives the wave spectrum integrated across all directions $E(\sigma) = \int F(\sigma, \theta) d\theta$ (how is directional distribution estimated?)

Other variables are then deduced from this spectrum, including H_s , $Tm01$ and $Tm02$ as described above.

We have observations available over a roughly 6 month period spanning from October 2021 to April 2022, which includes the period for which forecasts are available. A timeseries of the observed significant wave height from the forecast period is included in figure 3.10. The peak in wave height late on the 12th of March is clearly visible on the plot.

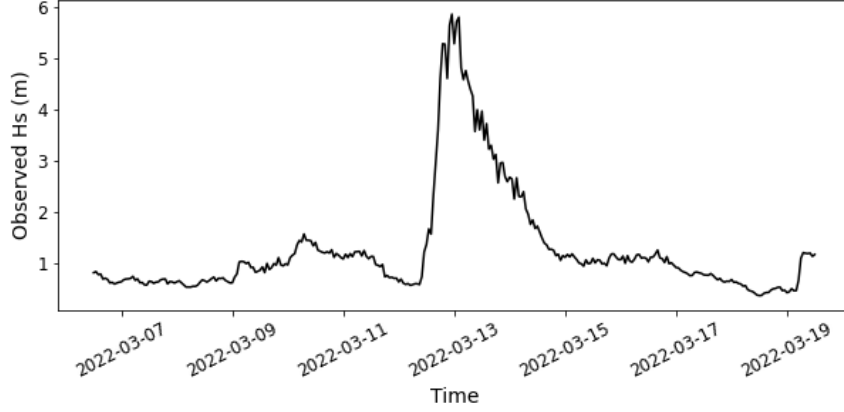


FIGURE 3.10: Observed timeseries of significant wave height from ADCP buoy

3.3.1 Diagnosis of Observation Errors

To simplify the computation, we assume that the observation error covariance matrix $\mathbf{R}$ is diagonal, essentially ignoring covariances between observations taken at different times. This also makes determining $\mathbf{R}$ easier since all that is required is to estimate the observation error variance at each point. To determine the magnitude of the observation errors, several methods are available. A simple one is to plot a histogram of observation - ensemble mean, called an $o - b$ plot. The spread in $o - b$ comes from a combination of observation and background error, so a simple way of estimating the observation error is to assume that observation error accounts for a constant proportion of this error, say 50%. An $o - b$ histogram for significant wave height is shown in figure 3.11. From the plot we see that the spread on the histogram is approximately 1m, so the standard deviation of observational errors in significant wave height may be estimated to be approximately 50cm.

Another thing that is visible from the $o - b$ plot is a bias in either the H_s forecasts or observations. The peak of the histogram is centred at around 30cm, which means that the forecast model tends to predict wave heights that are about 30cm less than those observed. If this bias is a result of the model, it would be desirable to investigate the capability of DA to reduce this bias. However, with the twin experiment method that is used in this project, this is not possible (see section 3.4 for details).

The histogram approach may be quantified more precisely following Desroziers et al., 2006, by defining $d_b^o = \mathbf{y} - \mathbf{H}\mathbf{x}_b$ and using the relation

$$\mathbb{E} \left[d_b^o (d_b^o)^\top \right] = \mathbf{H} \mathbf{P}^f \mathbf{H}^\top + \mathbf{R} \quad (3.2)$$

Taking traces and dividing by the number of observations, we get a good estimate for the observation error variance which is

$$\frac{\mathbb{E} \left[\sum_i (d_b^o)_i^2 \right]}{\# \text{ observations}} = \frac{\text{Tr}(\mathbf{H} \mathbf{P}^f \mathbf{H}^\top)}{\# \text{ observations}} \quad (3.3)$$

In our case, the observation operator is very simple and simply selects the forecast values from the appropriate location and times, so the diagonal elements of $\mathbf{H} \mathbf{P}^f \mathbf{H}^\top$

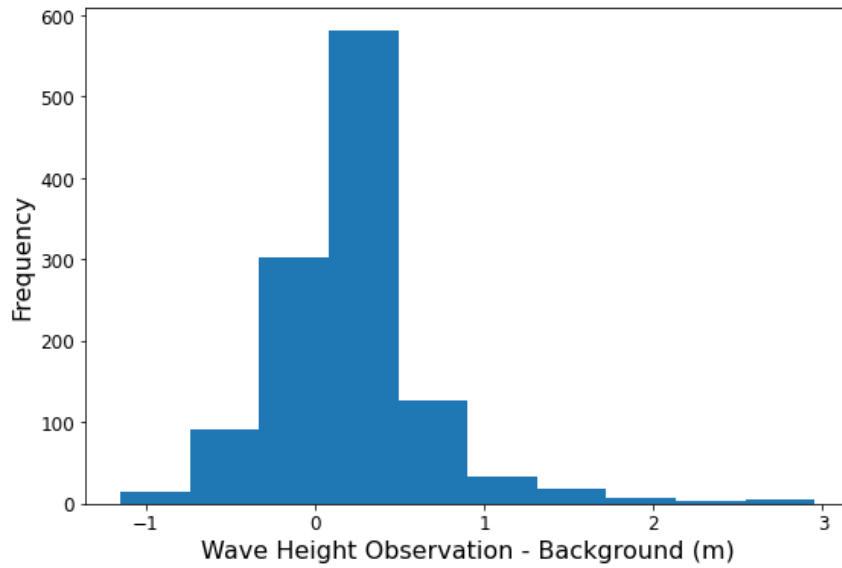


FIGURE 3.11: Observation - background histogram for significant wave height.

are simply the ensemble forecast variances at the observation location and times.

Another available method is that used in Houghton et al., 2023. They used a collocation study to estimate the error in wave height observations from a distributed network of drifting sofar spotter buoys. Pairs of simultaneous observations from locations within 50km of each other were collected and binned according to the average wave height of each pair. The difference of each pair was taken, and the standard deviation of differences from each bin was computed. They found a strong dependence of error on wave height.

In our case, we only have observations from one buoy and they are available hourly, so we take the pairs to be observations one hour apart. We then calculate the standard deviation using the method described above to obtain the plot in figure 3.12. As in Houghton et al., 2023, we find a strong linear relationship between errors and wave height, with the error standard deviation being approximately 9.6% of the wave height.

Recall from figure 3.6 that the ensemble variance H_s peaks just before the peak in H_s and drops back to lower values as H_s is at its highest value. Since ensemble variance is a good measure of the magnitude of the background error, the background error is large compared to observational error just before the peak. This means that observations from this period will be given higher weighting than most. During and shortly after the peak, the opposite is true: background error is small compared to observational error, so observations from this period will be given lower weighting.

Repeating this process with the observations of $Tm01$ and $Tm02$ yields the plots shown in figure 3.13. The error distributions for $Tm01$ and $Tm02$ also fall approximately on straight lines, though the fit is not as good as for H_s . The gradients are also shallower, with the error standard deviations being approximately 3.5% of the observed variable for both. So we will model the error standard deviation in H_s as being equal to 10% of H_s , and the error standard deviation in $Tm01$ and $Tm02$ as 3.5% of $Tm01$ and $Tm02$ respectively.

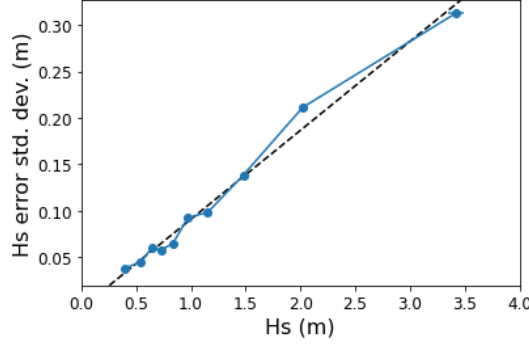


FIGURE 3.12: A plot of Hs error standard deviation against Hs.

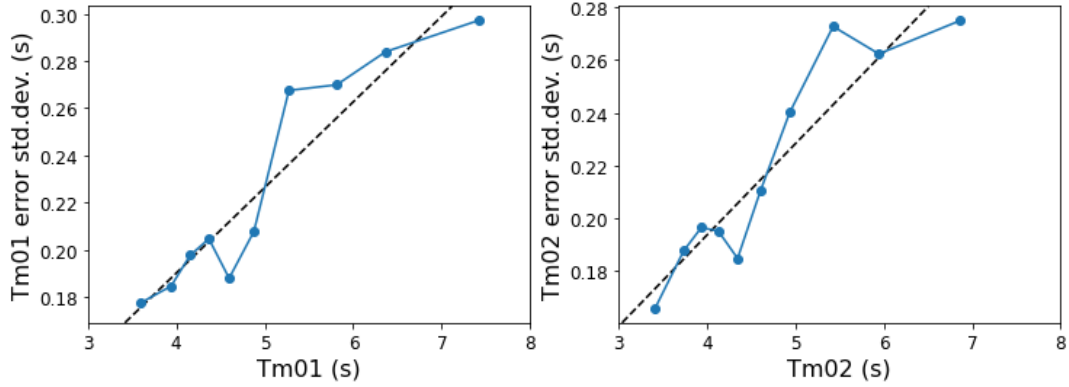


FIGURE 3.13: Plots of Tm01 and Tm02 error standard deviation against Tm01 and Tm02 respectively.

Since the errors in period are such a small percentage of the observations, it is possible that at low periods the error may become very small, which would lead the DA system fitting the observations too closely and producing physically unrealistic increments. To avoid this, we set a minimum error threshold ε_0 and set the observation error standard deviation equal to this threshold whenever the scaling method gives a value less than ε_0 . This threshold is determined using the method from Desroziers et al., 2006, using only the observations of wave height with a value of less than 5 seconds, as these are the observations which may lead to small errors.

The terms in 3.3 were calculated to be

$$\frac{\mathbb{E} \left[\sum_i (d_b^o)_i^2 \right]}{\# \text{ observations}} = 0.16s^2, \frac{\text{Tr}(\mathbf{H}\mathbf{P}^f\mathbf{H}^\top)}{\# \text{ observations}} = 0.11s^2$$

So the observation error variance is estimated to be $0.05s^2$ and the standard deviation threshold is set at $0.23s$.

3.4 Twin Experiment

In this study, we will not be using real observations in our assimilation but setting aside one ensemble member to represent the ‘truth’ and generating fictional observations based on this forecast data. This is known as a twin experiment, and has several advantages over assimilating real observations in this context:

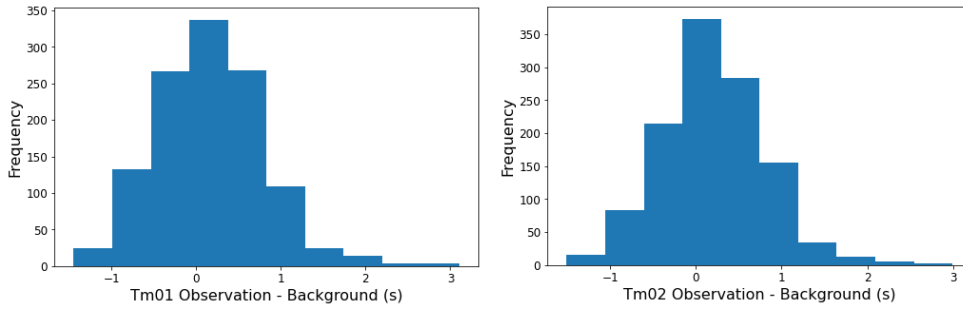


FIGURE 3.14: Observation minus background histograms for Tm01 (left) and Tm02 (right).

- The ‘truth’ is available away from the location of observations, so forecast error can be calculated over the whole model domain. This allows the effectiveness of DA to improve forecasts away from observation locations to be evaluated

The observations are generated by sampling values from the truth at predefined observation locations and times. Here we sample hourly values from the nearest gridpoint to the buoy to match the real observations that are available. To represent observation error, some random noise is added to the sampled values. This noise is drawn from a normal distribution with standard deviation equal to the error standard deviation derived in section 3.3

A downside of using the twin experiment method in this project is its inability to correct biases in the model. Figure 3.11 shows a possible bias in the forecasted values of H_s , which would be desirable to try to correct with DA. Since the data that we assimilate in this project is also from model output, it will be subject to all the same biases as the forecast data so it is impossible for the DA to correct these biases.

3.5 Data Assimilation Setup

To perform the data assimilation we use pyPDAF (**Chen2024**), which is a python interface for the fortran-based Parallel Data Assimilation Framework (PDAF). We will use the ETKF method which comes builtin with PDAF. As we are not able to run the WAVEWATCH III model ourselves, at each analysis step we will correct one 7-day forecast window all at once, and then the forecast step will consist of selecting data from the next 7-day window. In this way the time within the 7-day window is treated like another spatial coordinate by the DA system. A schematic of this process is included in figure 3.15

At each analysis step, we will assimilate observations from the first three days of the forecast window. This allows us to investigate errors at times after the assimilation window and assess the capability of DA to improve the forecast at future time when observations are unavailable.

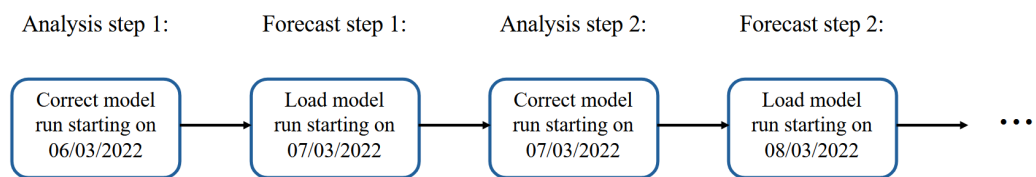


FIGURE 3.15: Schematic of the forecast and analysis steps from the ETKF scheme employed in this project.

Chapter 4

Results

This chapter contains an analysis of the errors in the DA output as compared to the model output with no DA. We also examine the results of assimilating Hs alone (Hs DA) compared to assimilating Hs, Tm01 and Tm02 (all vars DA).

For the purposes of a simple visual examination, we include timeseries of ensemble mean from each run, plotted alongside the truth. Figure 4.1 shows this for variables Hs and Tm01 at the buoy location from the 12Z 09/03/2022 model run. We see little change in Hs during the assimilation window as the no DA forecast matches the truth very closely here. After the end of the window, we see that Hs DA matches truth very well, while all vars DA is close to no DA. From the Tm01 plot, we see that Hs is similar to no DA during the window while all vars DA is close to the truth. After the window, the performance is more variable: Hs DA does well between 3 and 4 days, all runs match the truth closely between 4 and days, and none are very close after 5 days. (steps in no DA at 5-6 days? Ask Irina)

For comparison, figure 4.2 shows the same timeseries at 30° N 80° W, which is outside the GoM in the northern Atlantic ocean. As this location is so far from the buoy, we do not expect to see any kind of improvement from the DA, and indeed this is what we see. All vars DA, Hs DA and no DA are all a similar distance from the truth which indicates that DA has not improved the forecast. One thing to note is that there is often a large difference between the no DA line and both the Hs DA and all vars DA lines, meaning that the increments $\mathbf{d}_b^a = \mathbf{x}_a - \mathbf{x}_b$ are quite large. This is physically unrealistic as we would not expect the ocean state at the buoy location to have any effect on the ocean state at 30° N 80° W. This is explored further in section 4.5.

4.1 RMSE

Figure 4.3 shows the RMSE of the ensemble mean from no DA, Hs DA and all vars DA. The mean is taken over a 24-hour period from 12Z 12/03/2022 to 12Z 13/03/2022. This period was chosen because it contains the highest waves in the forecast period, which are associated with higher errors. (back this up) Forecasts of this period are available from multiple start dates, giving a range of leadtimes. Errors are plotted for Hs and Tm01. Tm02 is not included as it has a very similar form to Tm01.

For Hs, we see that during the assimilation window both Hs DA and all vars DA show an improvement of $\sim 60\%$ over no DA. After the end of the window, the errors from all runs increase and at 3-4 days leadtime, Hs DA shows a much greater RMSE reduction than all vars DA. At 4-5 days, Hs DA has similar error to no DA and all vars DA shows an improvement of $\sim 50\%$. 6-7 is similar except the improvement from

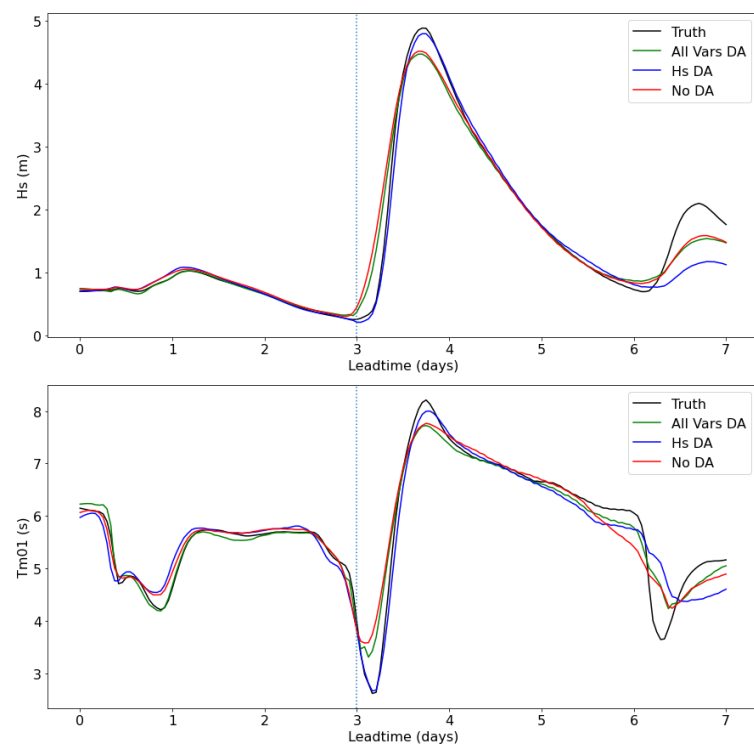


FIGURE 4.1: Timeseries of Hs (top) and Tm01 (bottom) at buoy location from 12Z 09/03/2022 model run. No DA ensemble mean is shown in red, Hs DA ensemble is shown in blue, all vars DA ensemble mean is shown in green and truth is shown in black. Blue dotted line indicates the end of the assimilation window.

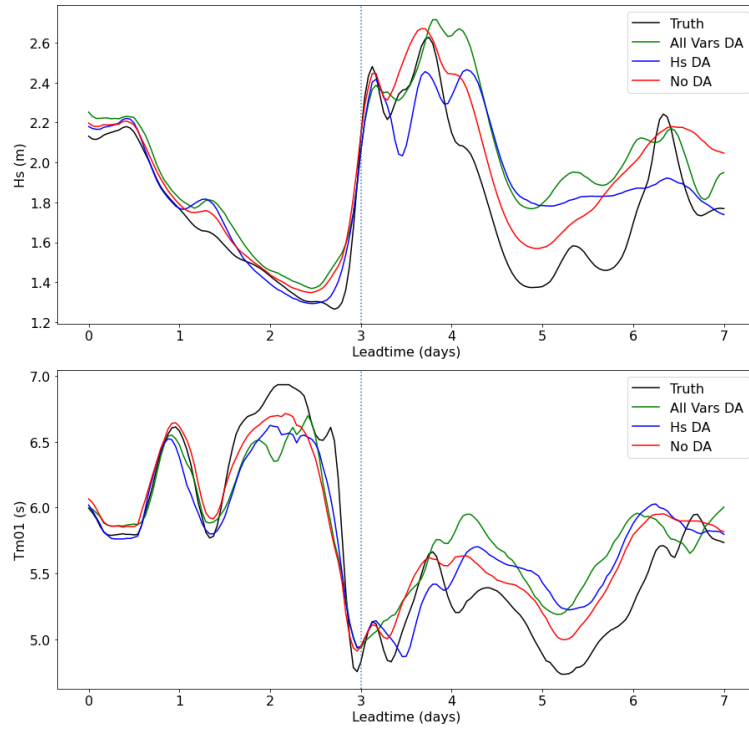


FIGURE 4.2: Timeseries of Hs (top) and Tm01 (bottom) at 30°N 80° W from 12Z 09/03/2022 model run. No DA ensemble mean is shown in red, Hs DA ensemble is shown in blue, all vars DA ensemble mean is shown in green and truth is shown in black. Blue dotted line indicates the end of the assimilation window.

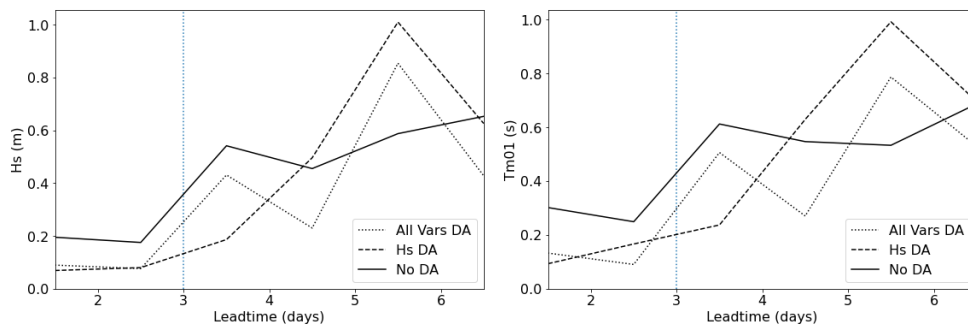


FIGURE 4.3: RMSE in Hs (left) and Tm01 (right) ensemble mean at the buoy location as a function of leadtime. RMSE is calculated over a 24hr period. Results shown for no DA (solid line), Hs DA (dashed line) and all vars DA (black dotted line). Blue dotted line indicates the end of the assimilation window.

all vars DA is around 35 %. At 5-6 days leadtime, both Hs DA and all vars DA have higher RMSE than no DA. (why)

For Tm01 the results are similar, with consistent improvement from DA within the assimilation window and more variable results afterwards. All vars DA still performs better than Hs DA with the exception of 1-2 days and 3-4 days leadtime. It is notable that Hs DA gives improvements in this case since no observations of Tm01 are actually assimilated. This can be explained by the fact that Hs and Tm01 are positively correlated (see figure 3.8). One difference between the errors in Tm01 and Hs and that Hs DA does slightly worse in predicting Tm01, when using the other two runs as a benchmark. This is unsurprising as we would expect assimilating Hs, Tm01 and Tm02 to give better predictions of Tm01 than Hs alone.

Figure 4.4 shows the spatial distribution of RMSE in Hs ensemble mean over the period from 12Z 12/03/2022 to 12Z 13/03/2022. At 2-3 days leadtime, both Hs DA and all vars DA show significant improvements in the immediate vicinity of the buoy. Hs DA shows RMSE reduction in the western GoM that is weaker than the reduction near the buoy, and the eastern GoM has increased RMSE. Outside the GoM there is very little change. All vars DA has more regions within the GoM where the RMSE is increased, and the increase in the eastern GoM is larger than that shown by Hs DA. There is also a noticeable reduction outside the GoM.

At 4-5 days leadtime, all vars DA shows reduced RMSE in the western GoM and increased RMSE further from the buoy in the eastern gulf. The change in RMSE outside the gulf is smaller relative to that seen within the GoM, but the magnitude of RMSE change over the whole domain is generally larger than that seen at 2-3 days. This is likely partly due to the fact that the no DA RMSE is larger, and partly related to the increments (expand and link). Hs DA shows smaller RMSE changes overall than all vars DA, and there is no significant reduction near the buoy.

Figure 4.5 shows the spatial distribution of RMSE in Tm01 ensemble mean over the period from 12Z 12/03/2022 to 12Z 13/03/2022. At 2-3 days leadtime, both Hs DA and all vars DA show improvements over no DA in the vicinity of the buoy with some increased errors in the eastern GoM and in the Atlantic. The general scale of error reduction/increase is the same in both cases. All vars DA also has regions of increased errors in the western GoM.

At 4-5 days leadtime, all vars DA shows improvement through most of the western GoM, with some increased errors again in the eastern GoM and in the Atlantic. Hs DA shows little change in RMSE through most of the domain, with the exception of a large increase in the northeastern GoM.

4.2 Correlation

4.3 Ensemble Spread

Ensemble spread is an important measure of uncertainty in ensemble forecasts. One aim of this project is to reduce uncertainty in the forecasts so that they may be used to inform decisions with greater confidence. Figure 4.6 shows the ensemble standard deviation in Hs and Tm01 at the buoy location from two different model runs. Both Hs DA and all vars DA show a reduction in ensemble spread in both variables. All vars DA shows a greater reduction in Tm01 ensemble spread overall. This is unsurprising since Tm01 is not directly assimilated in Hs DA. This means that any increment to

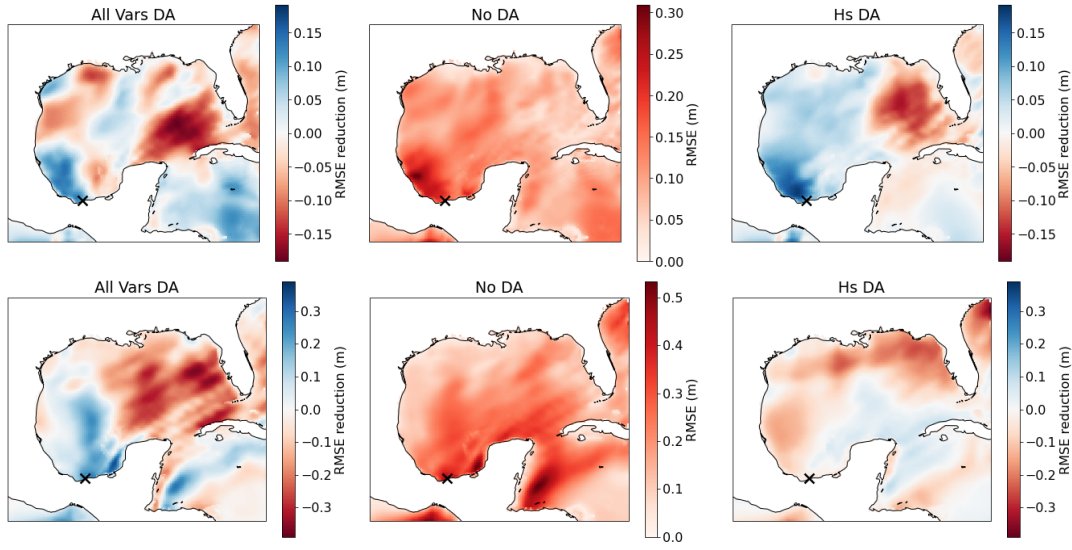


FIGURE 4.4: Maps of RMSE in Hs ensemble mean from the period 12Z 12/03/2022 to 13/03/2022, with leadtime 2-3 days (top) and 4-5 days (bottom). Middle panels show RMSE from no DA, side panels show absolute RMSE reduction compared to no DA from all vars DA (left) and Hs DA (right). The buoy location is marked with a cross.

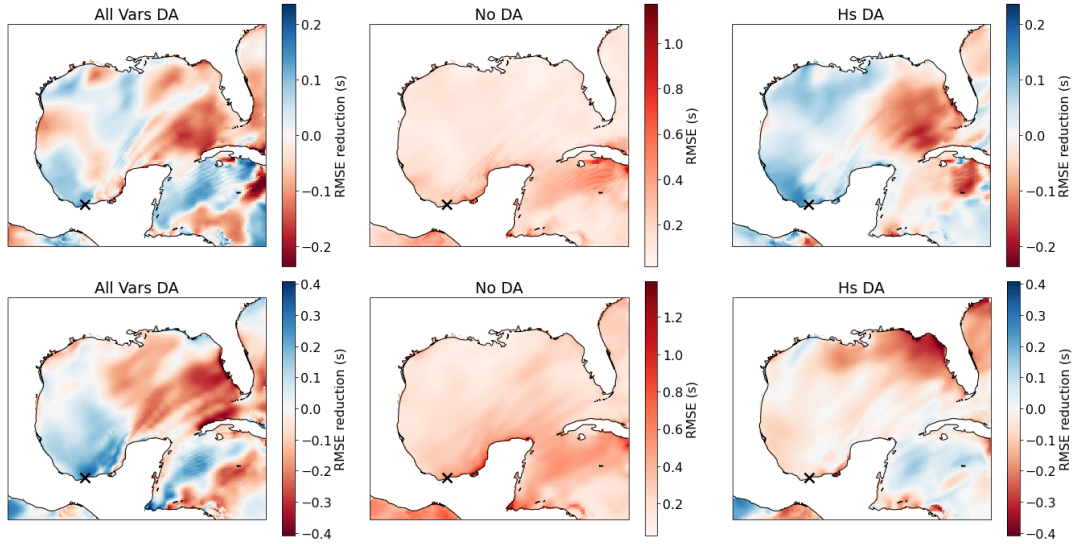


FIGURE 4.5: Maps of RMSE in Tm01 ensemble mean from the period 12Z 12/03/2022 to 13/03/2022, with leadtime 2-3 days (top) and 4-5 days (bottom). Middle panels show RMSE from no DA, side panels show absolute RMSE reduction compared to no DA from all vars DA (left) and Hs DA (right). The buoy location is marked with a cross.

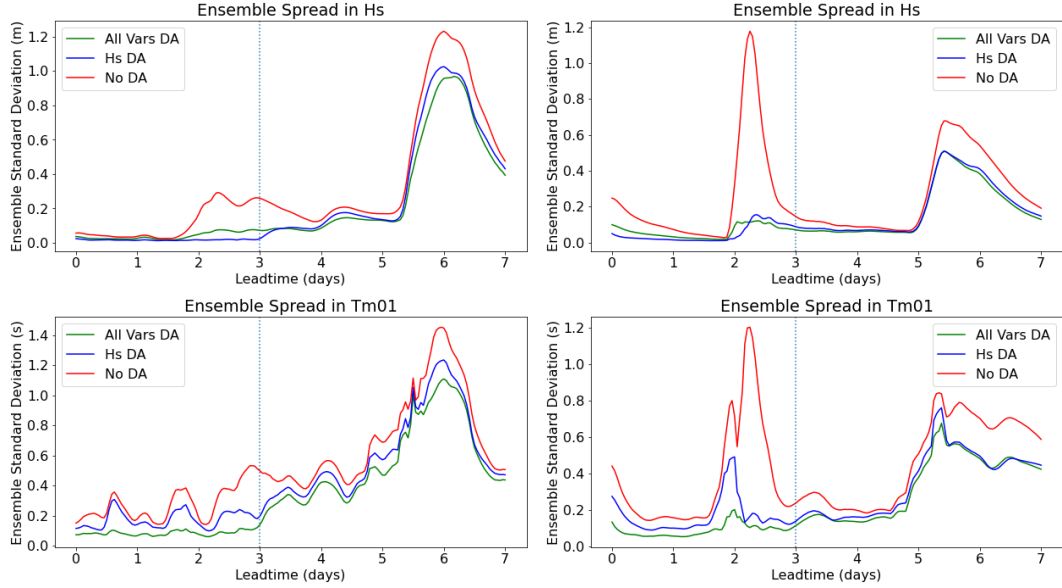


FIGURE 4.6: Timeseries of ensemble standard deviation in Hs (top) and Tm01 (bottom) at the buoy location. Taken from 12Z 06/03/2022 (left) and 12Z 10/03/2022 (right) model runs.

Tm01 from Hs DA comes from cross covariances between Hs and Tm01, which we generally expect to be small. In Hs ensemble spread, both runs perform similarly but Hs DA generally shows lower spread during the assimilation window and the opposite is true after the assimilation window. This is likely because the observational error scale in Hs was increased from 10% in Hs DA to 12% in all vars DA.

Figure 4.7 shows the spatial distribution of Hs ensemble spread at 12Z 11/03/2022, from 1 day and 5 days leadtime. At both leadtimes, All vars DA shows a greater reduction in ensemble spread than Hs DA across most of the domain. The only exception to this is the vicinity of the buoy at 1 day leadtime, when Hs DA shows a greater reduction. This is similar to what we saw in the timeseries, when Hs DA had greater spread reduction outside of the assimilation window. This reflects the fact that the dimensions of space and time within a forecast window are treated equivalently by our DA system. The greatest reductions in spread are where the spread of no DA is greatest.

Figure 4.8 shows the same as 4.7 but for Tm01. At 1 day leadtime, the pattern of reductions from both all vars DA and Hs DA closely match the pattern of standard deviation from no DA, and the reductions from all vars DA are stronger overall. At 5 days leadtime, the reduction from all vars DA generally matches the pattern of standard deviation from no DA, with a peak concentrated on the western side of the Yucatan peninsula. This peak is also present in Hs DA, which shows little reduction in spread elsewhere.

However, it would be short-sighted to focus solely on reducing the ensemble spread, as this may lead to the ensemble being overconfident. As described in section 2.1.7, it is detrimental to the forecast to have ensemble spread smaller than the forecast errors. To test whether this problem has arisen, we plot the ensemble standard deviation against

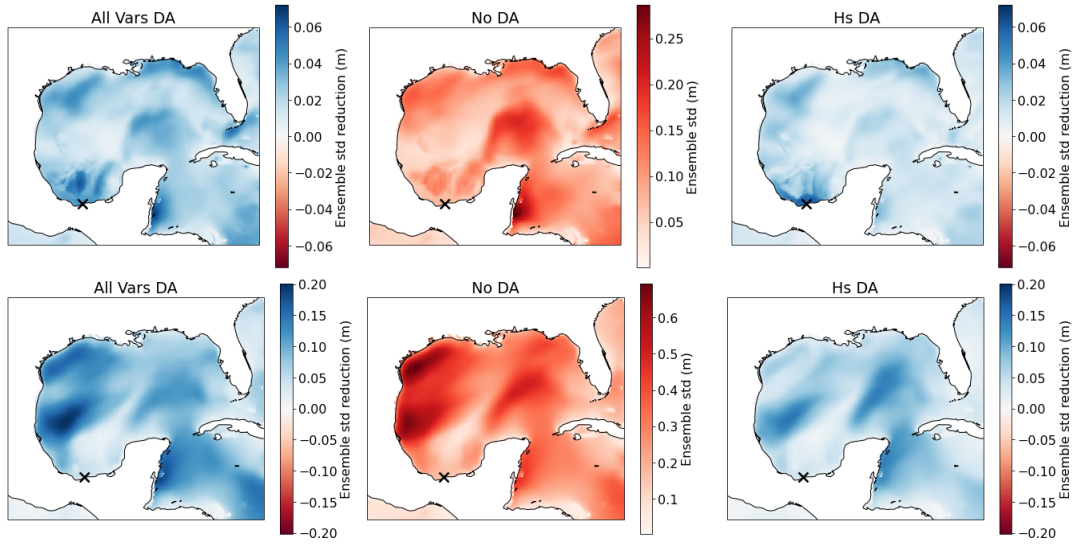


FIGURE 4.7: Ensemble standard deviation in Hs at 12Z 11/03/2022, taken from 12Z 06/03/2022 (top) and 12Z 10/03/2022 (bottom) model runs. Middle panel shows standard deviation from no DA, side panels show standard deviation from all vars DA (left) and Hs DA (right).

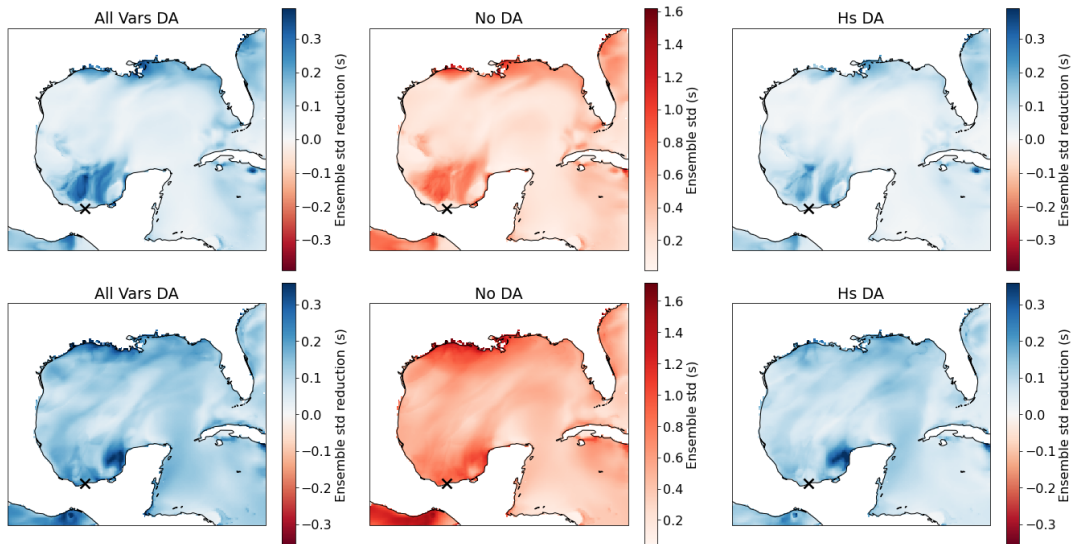


FIGURE 4.8: Ensemble standard deviation in Tm01 at 12Z 11/03/2022, taken from 12Z 06/03/2022 (top) and 12Z 10/03/2022 (bottom) model runs. Middle panel shows standard deviation from no DA, side panels show standard deviation from all vars DA (left) and Hs DA (right).

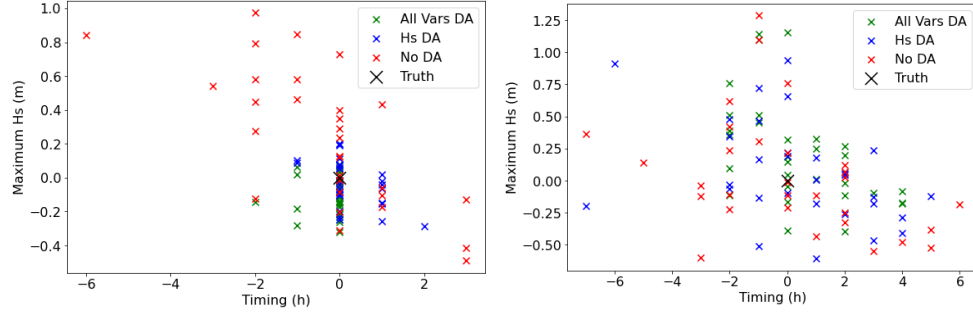


FIGURE 4.9: Scatter plots of timing and magnitude of maximum Hs at buoy location from period 12Z 12/03/2022 to 12Z 13/03/2022. Generated from model run starting 12Z 10/03/2022 (left) and 12Z 08/03/2022 (right). Truth is shown in black, and the no DA, Hs DA and all vars DA ensemble members are shown in red, blue and green respectively.

Leadtime 2-3 days			Leadtime 4-5 days		
	Timing (h)	Magnitude (m)		Timing (h)	Magnitude (m)
No DA	1.9	0.46	No DA	2.9	0.45
Hs DA	0.66	0.14	Hs DA	2.7	0.40
All Vars DA	0.51	0.15	All Vars DA	1.9	0.46

TABLE 4.1: RMSE in the timing and magnitude of maximum Hs as plotted in figure 4.9

4.4 Errors in Maximum Hs

Hs maxima are of particular interest in this project as they present the most danger to offshore operations. In this section we examine the effect of DA on the errors in the timing and magnitude of the Hs maximum on 12/03/2022. Figure 4.9 shows scatter plots of the timing and magnitude from no DA, Hs DA and all vars DA, relative to the truth. The maximum is selected from the period spanning 12Z 12/03/2022 to 12Z 13/03/2022. The left plot is generated from the model run starting 12Z 10/03/2022 and the right plot is generated from the model run starting 12Z 08/03/2022, giving leadtime ranges of 2-3 days and 4-5 days respectively. Table 4.1 shows the ensemble RMSE in timing and magnitude from each ensemble for both runs.

With a leadtime of 2-3 days, we see a large improvement in forecasts of timing and magnitude from both DA runs. From the scatter plot, the ensemble spread is considerably tighter around the true value, and the RMSE in both quantities is reduced. All vars DA does slightly better than Hs DA in predicting the timing of the maximum, and vice versa for the magnitude, but the differences between the two are small compared to the difference of both with no DA.

After 4-5 days, the errors are larger in all quantities but the magnitude from the no DA run. The differences between the two DA runs is larger, as Hs DA has a similar error in timing to no DA, but shows an improvement in the magnitude of 13%. Conversely, all vars DA has similar error in magnitude to no DA, but shows an improvement in timing of 35%

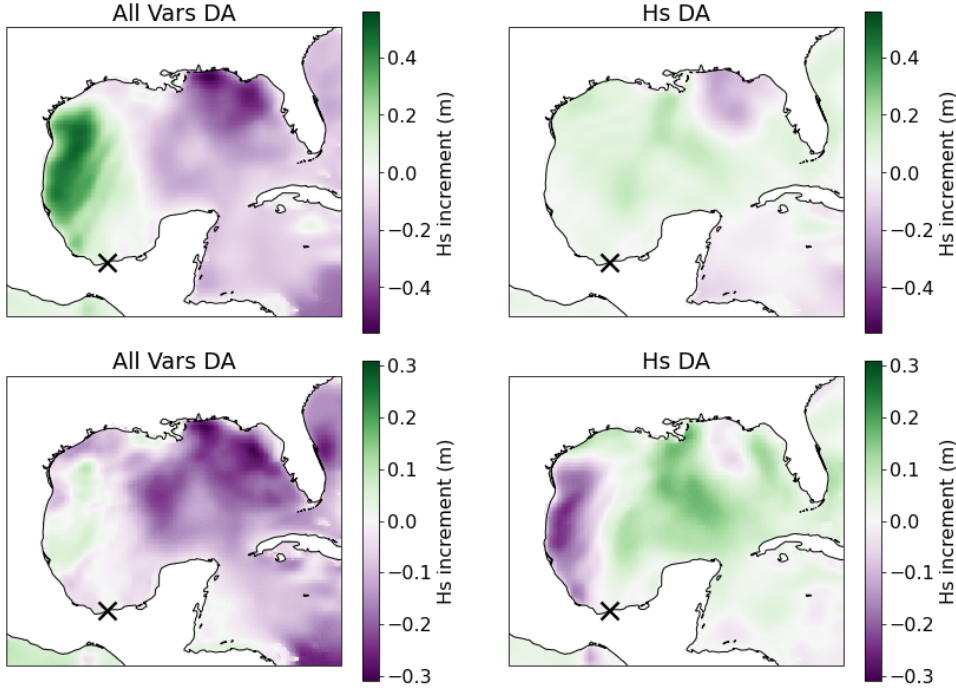


FIGURE 4.10: Hs increments at 12Z 15/03/2022 from all vars DA (left) and Hs DA (right). Taken from 12Z 10/03/2022 forecast run ensemble member 10 (top) and ensemble mean (bottom). The buoy location is marked with a cross.

4.5 Increments

The increments $\mathbf{d}_b^a = \mathbf{x}_a - \mathbf{x}_b$ are the corrections made to the background state by the DA system. From equation 2.14, the state that minimises the cost function will be close to the background when $\mathbf{B}$ is small compared to $\mathbf{R}$. This means that we expect the increments to be small in this situation. In general, we expect $\mathbf{B}$ to be small at locations far from the buoy and at times far from the assimilation window, since the covariance between ocean state at two points should generally decrease with increased distance/time separation.

Figure 4.10 shows the increments in Hs at 12Z 15/03/2022 taken from the 12Z 10/03/2022 forecast run. Increments are shown both for a randomly selected ensemble member and for the ensemble mean. We see that all vars DA has large increments far from the buoy location, which is physically unrealistic as described above. This could be due to the observation error being set too small, as this would lead the analysis state to deviate too far from the background in order to match the observations. This could be fixed with further refinement of the observation error scale. It could also be due to spurious long-distance covariances in $\mathbf{P}^f$, which might arise due to sampling error when calculating $\mathbf{P}^f$. This could be fixed with localisation.

Appendix A

Frequently Asked Questions

A.1 How do I change the colors of links?

The color of links can be changed to your liking using:

```
\hypersetup{urlcolor=red}, or
```

```
\hypersetup{citecolor=green}, or
```

```
\hypersetup{allcolor=blue}.
```

If you want to completely hide the links, you can use:

```
\hypersetup{allcolors=.}, or even better:
```

```
\hypersetup{hidelinks}.
```

If you want to have obvious links in the PDF but not the printed text, use:

```
\hypersetup{colorlinks=false}.
```


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